

# Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Grade 6 Mathematics · Standard 6.AT.6c

Name: \_\_\_\_\_

Date: \_\_\_\_\_

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

## Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

### Question 1 SELECTED RESPONSE · 1 POINT

Evaluate  $3 + 4 \times 2$ .

- ☐ A 9
- ☐ B 11
- ☐ C 14
- ☐ D 24

### Question 2 SELECTED RESPONSE · 1 POINT

What does  $2^3$  mean?

- ☐ A  $2 \times 2 \times 2$
- ☐ B  $2 \times 3$
- ☐ C  $2 + 2 + 2$
- ☐ D  $3 \times 3$

### Question 3 SELECTED RESPONSE · 1 POINT

Sofia buys 4 tickets at \$20 each and 2 tickets at \$15 each. Which expression shows the total?

- ☐ A  $4 \times 20 + 2 \times 15$
- ☐ B  $4 + 20 + 2 + 15$
- ☐ C  $(4 + 2) \times (20 + 15)$
- ☐ D  $4 \times 2 + 20 \times 15$

**Question 4**    SELECTED RESPONSE · 1 POINT

Evaluate  $5^2 + 3$ .

- ☐ A 13
- ☐ B 16
- ☐ C 28
- ☐ D 64

**Question 5**    SELECTED RESPONSE · 1 POINT

What is the total of  $2 \times 15 + 4 \times 20 + 8 \times 2^3$ ?

- ☐ A 126
- ☐ B 174
- ☐ C 210
- ☐ D 1,392

**Question 6**    SELECTED RESPONSE · 1 POINT

Miguel evaluates  $12 - 2 \times 3$  and gets 30. Margaret gets 6. Who is right?

- ☐ A Miguel, because he worked left to right
- ☐ B Both are right
- ☐ C Neither is right
- ☐ D Margaret, because multiplication comes before subtraction

**Part B — Test-Format Questions**

These match the state test's formats: two-part questions, select-all, and written responses.

**Question 7****TWO-PART QUESTION****Part A**

Evaluate the numerical expression  $3 \times 4^2 - 8$ .

*Fill in the bubble next to the one best answer.*

- ☐ (A) 16
- ☐ (B) 24
- ☐ (C) 40
- ☐ (D) 136

**Part B**

*Answer Part A before Part B — Part B asks about your reasoning.*

Which reasoning shows why your answer to Part A is correct?

- ☐ (A) Multiply  $3 \times 4$  first, then square the product:  $12^2 = 144$ , and  $144 - 8 = 136$ .
- ☐ (B) Evaluate the power first:  $4^2 = 16$ . Then multiply:  $3 \times 16 = 48$ . Then subtract:  $48 - 8 = 40$ .
- ☐ (C) Read  $4^2$  as  $4 \times 2 = 8$ , then multiply  $3 \times 8 = 24$  and subtract:  $24 - 8 = 16$ .
- ☐ (D) Subtract first because 16 and 8 are next to each other:  $16 - 8 = 8$ , then  $3 \times 8 = 24$ .

**Question 8****WRITTEN RESPONSE · 2 POINTS****Type II Reasoning: Subtracting Too Soon**

Devon evaluated  $20 - 2 \times 3^2$ . His work was:  $3^2 = 9$ , then  $20 - 2 = 18$ , then  $18 \times 9 = 162$ . He wrote that the value of the expression is 162.

Explain Devon's misconception and give the correct value of the expression.

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

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## Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

### Question 9 WRITTEN RESPONSE · 2 POINTS

Seven friends split the bill for a pizza party. They order 4 pizzas at  $3^2$  dollars each, plus a one-time \$6 delivery fee. Write ONE numerical expression for what each friend owes, evaluate it, and explain why the grouping symbols matter.

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

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When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.